Technical Document #345: Blockchain - Comprehensive Overview

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Non-fungible tokens (NFTs) represent unique digital assets on the blockchain. They have transformed digital ownership and art markets.

Proof of Stake (PoS) selects validators based on their stake in the network. It is more energy-efficient than Proof of Work.

Tokenomics studies the economic systems of cryptocurrency tokens. It covers supply mechanisms, incentive structures, and value capture.

Zero-knowledge proofs allow one party to prove knowledge of information without revealing the information itself. They enhance privacy on blockchains.

Decentralized finance (DeFi) recreates traditional financial systems using blockchain technology. It enables lending, borrowing, and trading without intermediaries.

Sustainability and environmental responsibility are becoming key considerations in technology design. Green computing aims to reduce the environmental impact of technology.

Edge computing processes data closer to its source, reducing latency and bandwidth usage. It is essential for real-time applications like autonomous vehicles.

Remote work technologies have transformed the workplace. Collaboration tools and cloud services enable distributed teams to work effectively from anywhere.

Cybersecurity threats continue to evolve in complexity and scale. Organizations must invest in robust security measures to protect sensitive data.

Key Takeaways:

- Cross-chain technology enables communication between different blockchains.
- Blockchain is a distributed ledger technology that records transactions across many computers.
- Proof of Work (PoW) is a consensus mechanism where miners solve complex puzzles to validate transactions.